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Studierichting: Informatica
Oefeningenreeks 3D nr. 7.2

$f(x) = e^x \sin x$

$f'(x) = e^x \sin x + e^x \cos x = e^x (\sin x + \cos x)$

$f'(x) = 0 \Leftrightarrow e^x = 0 \vee \sin x + \cos x = 0$

$e^x$:

$e^x = 0$ kan niet, $e^x$ raakt nooit y = 0.

$\sin x + \cos x$:

$\sin x + \cos x = 0$

$\Leftrightarrow \sin x = -\cos x$

$\Leftrightarrow \dfrac{\sin x}{\cos x} = -1$

$\Leftrightarrow \tan x = -1$

$\Leftrightarrow x = -\dfrac{\pi}{4} + k\pi$

We checken waar de minima en maxima zitten:

$-\dfrac{\pi}{4}$:

$-\dfrac{\pi}{2} < -\dfrac{\pi}{4} \Rightarrow f'\left(-\dfrac{\pi}{2}\right) = -e^{-\frac{\pi}{2}} < 0$

$0 > -\dfrac{\pi}{4} \Rightarrow f'(0) = 1 > 0$

| $x$ | $-\infty$ | | $-\frac{\pi}{2}$ | | $-\frac{\pi}{4}$ | | $0$ | | $+\infty$ |
|---|---|---|---|---|---|---|---|---|---|
| $f'(x)$ | | $-$ | $-$ | $-$ | $0$ | $+$ | $+$ | $+$ | |
| $f(x)$ | | $\searrow$ | $\searrow$ | $\searrow$ | $m$ | $\nearrow$ | $\nearrow$ | $\nearrow$ | |

$x = -\dfrac{\pi}{4} + k2\pi$ zijn lokale minima

$\dfrac{3\pi}{4}$:

$0 < \dfrac{3\pi}{4} \Rightarrow f'(0) = 1 > 0$

$\pi > \dfrac{3\pi}{4} \Rightarrow f'(\pi) = -e^{\pi} < 0$

| $x$ | $-\infty$ | | $0$ | | $\frac{3\pi}{4}$ | | $\pi$ | | $+\infty$ |
|---|---|---|---|---|---|---|---|---|---|
| $f'(x)$ | | $+$ | $+$ | $+$ | $0$ | $-$ | $-$ | $-$ | |
| $f(x)$ | | $\nearrow$ | $\nearrow$ | $\nearrow$ | $M$ | $\searrow$ | $\searrow$ | $\searrow$ | |

$x = \frac{3\pi}{4} + k2\pi$ zijn lokale maxima

# References